GG-B-406a
JANUARY 28, 1958
SUPERSEDING
Int. Fed. Spec. GG-B-00406 (GSA-FSS)
March 28, 1957, and
Fed. Spec. GG-S-149
November 1, 1946

FEDERAL SPECIFICATION

BLADES AND FRAME, DENTAL LABORATORY SAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers an adjustable saw frame and blades suitable for general dental use for cutting plastics and gypsum products.

1.2 Classification.

1.2.1 *Type and classes.*—This specification covers one type of saw frame, and two classes of blades, as specified (see 6.2).

Class 1.—Flat blade.

Class 2.—Spiral blade.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-585—Boxes; Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Lincers, Case, Waterproof.

MIL-P-12902 — Packaging of Surgical Dental, and Veterinary Instruments, Diagnostic Equipment and Related Items.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material shall be as hereinafter specified. When materials are used which are not specifically designated, they shall be of the quality best suited for the purpose intended.

3.1.1 *Steel.*

3.1.1.1 *Frame.*—Saw frames shall be of steel rod, heat treated to produce frames conforming to the requirements specified herein.

3.1.1.2 *Blades.*—Steel used in the manufacture of blades shall be good quality saw steel having a carbon content of 0.7 to 0.8 percent. Steel shall be heat treated for cutting plastics and gypsum products. Blades shall be treated to retard corrosion.

3.1.2 *Hardwood.*—Handle for frames shall be made of hardwood suitable for the purpose, and shall be thoroughly seasoned.

3.2 Frame.

3.2.1 *Design and construction.*—Saw frame shall be of the adjustable type to accommodate blades measuring 3 to 6 inches, inclusive, in length, and so constructed to hold the blade securely and maintain rigidity during use. Frame shall have a minimum depth clearance of 2½ inches. Depth shall be measured from teeth of blade to nearest inside edge or projection of frame. Frame shall be so designed that all parts of the frame shall be held together when the blade is not in place. Style and design of frame shall be as shown in figure 1. The illustration is not restrictive and is not intended to preclude the purchase of frames otherwise conforming to this specification. Frame shall consist essentially of a bow, bow support and tang, and a handle as hereinafter specified:

3.2.1.1 *Bow.*—The bow shall be made of 0.240 ± 0.015 inch round steel rod, 9 ± ¼ inches in overall length, bent downward at 7 ± ¼ inches from the end to form a right angle. The top portion of the rod shall be grooved for a distance of not less than 3¼ inches from the free end. The front end of the bow shall terminate in a blade holding device. The blade-bearing surface of the holding device shall be smoothly finished to provide a flat bearing surface for the blade.

3.2.1.2 *Bow support and tang.*—The upright portion of the tang shall be rectangular in shape with a receptacle at the top for holding the bow. A clamping device shall be provided with this receptacle for securing the

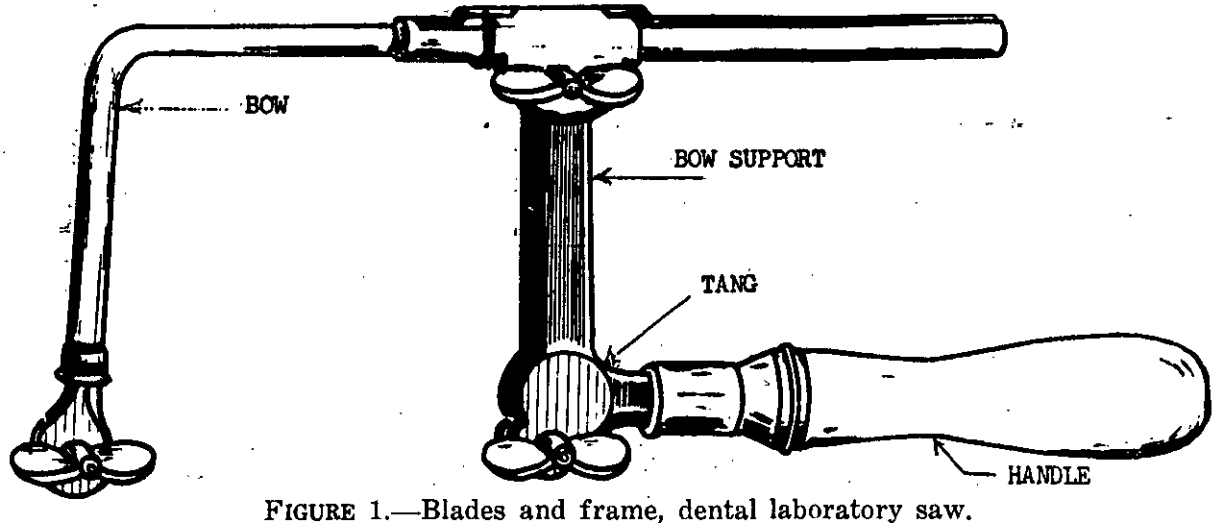


FIGURE 1.—Blades and frame, dental laboratory saw.

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bow and for adjusting to the desired length. A brass key furnished with the receptacle shall fit snugly in the groove in the bow. The bottom of the upright shall have the same type of blade-holding device as on the front end of the bow. The tang shall extend at a right angle to the upright and shall provide an attachment for the handle.

3.2.1.3 *Handle.*—Handle shall be shaped to provide a comfortable handgrip. Handle shall be made of wood as specified in 3.1.2, shall be smoothly finished, and lacquered or suitably painted. It shall be 4 ± ¼ inches in length and ⅞ inch at the maximum diameter. Handle shall be securely attached to the tang by means of a heavy tubular steel ferrule, ½ inch in diameter.

3.3 Finish.—All steel parts of the frame shall be either nickel or chromium over nickel plated. When subjected to the boil test specified in 4.3.1, the total surface area showing corrosion after testing shall not exceed 2 percent.

3.4 Saw blades.

3.4.1 *Class 1, flat.*—Blades shall have a single cutting edge and shall be designed with flat blank ends for clamping into the jaws of the frame. Blades shall conform to the dimensions shown in Table I, and shall have 13 to 15 teeth per inch.

3.4.2 *Class 2, spiral.*—Blades shall be spiral in design, so as to cut in any direction without turning the frame. Blades shall have a minimum of eight twists to an inch for the cutting length. Blades ends shall be designed for clamping into the jaws of the frame. Blades shall conform to the dimensions shown in table I, and shall have 33 to 39 teeth per inch.

3.5 Identification marking. — The frame shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.6 Workmanship.—Workmanship shall be first class throughout. Frames shall be free from fins, burrs, external sharp or rough edges, corners, or surfaces, and defects which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standards MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at Government laboratory.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be disovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified

TABLE I.—*Dimension of blades*

Class	Nominal Length (inches)	Length Of Cut (min.) (inches)	Thickness[1]			Blade[2] Width (inch)
			Back (inch)	Ends (inch)	Over Teeth (inch)	
1	5	3.5	0.011	--	0.013	0.022
2	5	3.1	--	0.016	--	0.035

[1] Plus or minus 0.001 tolerance allowable.
[2] Plus or minus 5 percent allowable.

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(see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.

4.3.1 *Boil test.* — Before testing, frames shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the frames shall be boiled in distilled water in a suitable glass receptacle for 30 minutes and allowed to remain submerged for 24 hours.

5. PREPARATION FOR DELIVERY

5.1 Unit of issue.

5.1.1 *Stock No. 6520–553–8150.*—One (1) frame as specified constitutes one unit.

5.1.2 *Stock No. 6520–553–8250.* — Twelve (12) blades as specified constitutes one unit.

5.2 Packaging quantities.—The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE II.—*Packaging quantities*

Federal Stock Number	Unit Package	Intermediate Package	Shipping Container
6520–553–8150	1 unit	6 units	96 units
6520–553–8250	1 unit	12 units	288 units

5.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.3 Stock No. 6520–553–8150.

5.3.1 *Packaging and packing.*

5.3.1.1 *Level A.*—Packaging and packing shall conform to Procedure No. 35 of Military Specification MIL–P–12902.

5.3.1.2 *Level B.*—Packaging and packing shall conform to Procedure No. 21 of Military Specification MIL–P–12902.

5.3.1.3 *Level C.*—The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3.2 *Marking.*—Marking shall conform to Military Specification MIL–P–12902, except that ASMPA stock number will not be required.

5.4 Stock No. 6520-553-8250.

5.4.1 *Packaging.*

5.4.1.1 *Level A.*

5.4.1.1.1 *Cleaning.*—Each blade shall be thoroughly cleaned of all shop dirt, grit, or other contamination.

5.4.1.1.2 *Preservation.* — Each blade shall be coated with a suitable rust-preventive compound.

5.4.1.1.3 *Unit package.* — Each unit (12 blades) shall be wrapped in anti-corrosive greaseproof paper and packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening.

5.4.1.1.4 *Intermediate package.*—Intermediate package shall be a carton of suitable size and design. Carton shall be adequately secured to prevent accidental opening.

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5.4.1.2 *Level C.*—Unless otherwise specified, commercial packages are acceptable.

5.4.2 *Packing.*

5.4.2.1 *Level A.*

5.4.2.1.1 *Exterior container.* — Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, Style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.4.2.1.2 *Case liner.* — Each level A box shall be lined with a waterproof case liner, conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.4.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.4.2.2 *Level B.*

5.4.2.2.1 *Exterior container.* — Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.4.2.3 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4.3 *Marking.*

5.4.3.1 *Unit package.*—Each unit envelope shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.4.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.4.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standards MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Class of blade required (see 1.2.1).
(*c*) Inspection required (see 4.2).
(*d*) Place of final destination (see 4.2).
(*e*) Selection of applicable levels of packaging, packing and marking (see 5.3.1, 5.4.1, and 5.4.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

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6.4 Federal specifications do not include all types, grades, classes, sizes, etc., of the commodities indicated by the title of the specification, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 This specification covers the following items listed in the Armed Services Medical Stock List:

6520–553–8150 — FRAME, DENTAL LABORATORY SAW

6520–553–8250 — BLADE, DENTAL LABORATORY SAW, Spiral, No. 2 Blade, 12s.

6.6 **Transportation description.**—Transportation description applicable to these items is:

Saw Blades, NOIBN:
Carload minimum weight 30,000 pounds.
Saw Blades:
Motor volume minimum weight 30,000 pounds.

Saw Frames, Tools, NOIBN:
Carload minimum weight 30,000 pounds.
Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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